Video Transcript

Video 4: Introduction to DASK and Running Parallel Operations (09:28)

We're going to start using DASK for parallel computing. Let's take a look first at some simple tasks and timing them. So, for example, let's suppose we want to time the execution in this cell. We can put in time, and let's do from time. Let's import sleep, and let's have a couple of functions. So, let's define increment, and we'll pass in an argument. And let's just sleep for one second, and return the result, x + 1. Okay, and let's do add, and add will add x and y. And again, we'll put in sleep for one second just to exaggerate the timing that each of these takes to execute. And, we'll return x + y.

Let's say x equals increment and we'll pass in one, and y equals increment, pass in two, and z equals add x and y, okay. And, we'll print z out. Okay, let's execute this, and we'll see that this takes three seconds. And it takes three seconds because to calculate the first x here, this increment call will sleep for a second, this one will sleep for a second, and this one again will sleep for one second. But, we could do these x calculation and y calculation in parallel, they're independent. So, now let's import from DASK import delayed.

And what we're going to do is we're going to put wrappers around these. So, let me copy those. We're going to put the decorator around this delayed, and we'll see exactly how it works, but it's going to automatically parallelize for us this calculation, delayed.

Okay so, let's calculate x equals delayed, and we're going to pass in increment and pass into it a value. And y equals delayed, pass in the function and the value two, say. And now z equal, let's call add this time, and here we'll pass in x and y. And now, we need to call compute because this actually won't do anything. Let me execute it and show you, let's put a timer in here. So, let's time this. So, it took one millisecond, and that's because this hasn't actually done the calculation yet. It's just setting it up.

And what do I mean by setting it up? Well, if we put in visualize, we'll see that. So, let me take out time here and run it with visualize. And we see that it shows us the graph of how it's going to do this calculation. So, it does the increments in parallel, and then, we'll go to the add. And, let's

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take a look at the time for it because what do you guess the time should be if it's doing the increments in parallel? But, the add. Now, we have to be careful because it hasn't actually done the calculation. To do the calculation, we need to tell it to compute. Let's run it, and you see that it takes two seconds.

So, before it took three seconds; now it's doing things in parallel. So, let's take a look at another operation that takes time. So, let's suppose we've got an array, and we've got a for loop that we're going to operate on the array. We'll still have the increment, so let's put that back. And now, let's have a for loop, for i in data. So, we're looping over that. Let's get ourselves a result array. There it is, and now, we're going to append to that. We'll increment i, so we've got that now, result, and let's get the total as the sum of the result. Okay, so let's see how long this takes. So, it took six seconds. So, each call in the for loop to increment took one second. Now, we can do better than that because we can do things in parallel. So, let's go back and we'll add our decorator. And now, here in result, we'll add that. And in this, let's delay this as well. And passing result, and let's print out the total. Okay so, this should now be all in parallel. Let's see what our timing is now. Remember, we didn't execute anything yet. We've got to do compute. So, three seconds.

So, this iteration, the whole iteration only took one second. But this delayed in the result, probably took a second as well. Now, let's take a look what was actually going on. Let's visualize it and see what it did. And we need to take this out. So, this is what actually happened in the computation. All of these were done in parallel, went up to the sum, and then that was the total. So, DASK does an amazing job of parallelizing operations, and we're going to see that later when we're operating on big data in lots of files. But that's an introduction to DASK and actually to the visualization. It's using graphviz to do this visualization. It's very neat indeed. Okay, bye for now.

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